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Direct-Reading Inhalable Dust Monitoring—An Assessment of Current Measurement Methods

ANDREW THORPE* and PETER T. WALSH

Health and Safety Laboratory, Harpur Hill, Buxton, Derbyshire SK17 9JN, UK

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Direct-reading dust monitors designed specifically to measure the inhalable fraction of airborne dust are not widely available. Current practice therefore often involves comparing the response of photometer-type dust monitors with the concentration measured with a reference gravimetric inhalable sampler, which is used to adjust the dust monitor measurement. However, changes in airborne particle size can result in significant errors in the estimation of inhalable concentration by this method. The main aim of this study was to assess how these dust monitors behave when challenged with airborne dust containing particles in the inhalable size range and also to investigate alternative dust monitors whose response might not be as prone to variations in particle size or that could be adapted to measure inhalable dust concentration. Several photometer-type dust monitors and a Respicon TM, tapered element oscillating microbalance (TEOM) personal dust monitor (PDM) 3600, TEOM 1400, and Dustrak DRX were assessed for the measurement of airborne inhalable dust during laboratory and field trials. The PDM was modified to allow it to sample and measure larger particles in the inhalable size range. During the laboratory tests, the dust monitors and reference gravimetric samplers were challenged inside a large dust tunnel with aerosols of industrial dusts known to present an inhalable hazard and aluminium oxide powders with a range of discrete particle sizes. A constant concentration of each dust type was generated and peak concentrations of larger particles were periodically introduced to investigate the effects of sudden changes in particle size on monitor calibration. The PDM, Respicon, and DataRam photometer were also assessed during field trials at a bakery, joinery, and a grain mill. Laboratory results showed that the Respicon, modified PDM, and TEOM 1400 observed good linearity for all types of dust when compared with measurements made with a reference IOM sampler; the photometer-type dust monitors on the other hand showed little correlation. The Respicon also accurately measured the inhalable concentration, whereas the modified PDM underestimated it by ~27%. Photometer responses varied considerably with changing particle size, which resulted in appreciable errors in airborne inhalable dust concentration measurements. Similar trends were also observed during field trials. Despite having limitations, both the modified PDM and Respicon showed promise as real-time inhalable dust monitors.

Keywords: direct reading; inertial impactor; inhalable; inhalable monitor; photometer; real time; tapered element

INTRODUCTION

Direct-reading dust monitors used for real-time measurements of respirable airborne dust concentration are widely available from a number of manufacturers. They are becoming an attractive and affordable alternative to conventional gravimetric methods because of their ability to identify and measure short-term fluctuations in airborne dust concentration. They typically find use in walk-through surveys, site dust measurements, assessment of the effectiveness of dust control systems, and measurement of indoor air quality (Willeke and Baron, 2001; Rosén *et al.,* 2005).

Most direct-reading dust monitors are of the light scattering (photometer) type where a collimated light source (laser or solid-state diode) illuminates dust entering the instrument and the intensity of

*Author to whom correspondence should be addressed. Tel: 01298218527; Fax: 01298218395; e-mail: Andrew.Thorpe@hsl.gov.uk

light scattered in the forward direction is proportional to the respirable (CEN, 1993) fraction of the airborne dust concentration. Physical properties of the dust such as colour, shape, and refractive index affect the instrument response but can be minimized by utilizing a narrow angle of forward scatter (Willeke and Baron, 2001). However, exposure to the inhalable (CEN, 1993) fraction of airborne dust is often of concern because it can cause serious health problems. A direct-reading dust monitor capable of measuring the inhalable fraction would therefore be a valuable tool for the occupational hygienist.

Light scattering dust monitors are usually calibrated in the factory using a 'standard' test dust and are adjusted to agree with respirable dust concentration measurements made using reference methods (HSE; MDHS 14/3, 2000). However, in order to make more accurate measurements, dust monitors should always be compared with a reference gravimetric sampler to determine an average calibration factor (CF) for the aerosol being measured. Typically, this would be a cyclone sampler (Higgins and Dewell, 1967) to measure the respirable fraction, although inhalable samplers such as the IOM sampler (Mark and Vincent, 1986) are also used in practice to calibrate photometer-type dust monitors to make real-time measurements of airborne inhalable dust.

However, extreme care should be taken when using photometer-type dust monitors to measure aerosols containing a significant fraction of inhalable-sized particles because the monitor response decreases significantly with particle sizes larger than the respirable size range (Thorpe, 2007). Consequently, small changes in the ratio of respirable to inhalable dust concentration during measurement may result in sizable errors. In such instances, the application of an average CF could easily result in an over- or underestimation of airborne concentrations. More knowledge is required to quantify the uncertainty of such calibrations.

Alternative dust monitors are available that might not be as prone to such errors or could be adapted to measure inhalable dust concentration in real time, such as:

- A personal dust monitor (PDM) based on tapered element oscillating microbalance (TEOM) technology;
- A multistage virtual impactor that simultaneously measures the three main occupation health-related size fractions with photometers built in to each stage to give time-resolved measurements of each size fraction;
- A light scattering dust monitor, which simultaneously measures the occupational and environmental health-related fractions of airborne dust by combining photometry with particle counting techniques.

The first aim of this study was to carry out laboratory measurements to investigate how currently available photometer direct-reading dust monitors behave when they are used to determine the concentration of airborne inhalable dust, and to evaluate other types of dust monitor for potential use as personal direct-reading inhalable monitors.

The second aim was to take a selection of direct-reading dust monitors on site to see if they could be used to accurately measure in real-time, airborne inhalable dust concentration throughout actual workplace activities.

MATERIALS AND METHODS

Measurement strategy

The performance of the dust monitors was assessed by comparing their responses with measurements made with reference IOM gravimetric samplers (Mark and Vincent, 1986; HSE, 2000) used to measure the reference inhalable airborne dust concentration during laboratory and field trials. During the laboratory trials, additional gravimetric cyclone samplers were also included with the intention of comparing dust monitor response with the respirable fraction.

A modified portable TEOM dust monitor (PDM 3600) and Respicon TM dust monitor were used as secondary standards for measuring inhalable dust concentration in real time and were used throughout the laboratory tests and field trials to investigate the effects of sudden changes in particle size on the response of photometer-type dust monitors. These are described in detail later.

Personal direct-reading monitors

Each photometer-type dust monitor was of the light scattering type and as such was most sensitive to particles in the respirable size range (typically 0.3–10 μm) and had data logging capabilities for subsequent data analysis. Tests on a selection of both personal and static dust monitors were carried out during this study

Photometer-type instruments.

Thermo Fisher DataRam (model pDR-1000): The DataRam relies on ambient air movement to introduce particles into the sensing chamber. It has a measurement range of 0.001–400 mg m^{-3} and is factory calibrated to measure 'SAEJ726 Fine' standard test dust, also known as 'Arizona Road Dust' (ISO 12103-1, 1997).

TSI Sidepak AM510: The Sidepak uses a built-in pump to introduce aerosol into its sampling inlet. It has a measurement range of 0.001–20 mg m^{-3} and is factory calibrated using 'SAEJ726 Fine' standard test dust. For the purpose of these tests, an IOM inlet adaptor was made so that only the inhalable size fraction was sampled. This consisted of a standard IOM inlet attached to the end of a pipe that had been bent through 90° and that was connected to the inlet nozzle on the Sidepak. Particle losses into the inlet were minimized by optimizing the pipe diameter, bend radius, and tube length (Willeke and Baron, 2001). In practice, the loss of large particles will have negligible effect because the instrument only responds to particles principally within the respirable size range (0.3–10 μm).

Casella Microdust Pro: The Microdust Pro is a portable dust monitor with a detachable wand, which is non-pumped and relies on ambient air movement or movement of the wand through the air to introduce dust into the sensing zone. It has a measurement range of 0.001–2500 mg m^{-3} and is calibrated against total suspended particulate (TSP) using a test aerosol of 'SAEJ726 Fine' standard test dust. The 'factory' calibration setting can be checked at any time using an optical filter supplied with the instrument.

Other real-time instruments.

Hund Respicon TM: The Respicon TM is a personal monitor, which simultaneously determines the inhalable, thoracic, and respirable size fractions by using a sampling head comprising a multistage, virtual impactor (Willeke and Baron, 2001) that collects airborne particles onto three collection filters. The first impactor stage separates out and collects particles smaller than 4 μm. The second stage collects particles below 10 μm, whereas the third stage collects the remaining coarse particles. The TM model also measures the three size fractions in real time by combining inertial classification with light scattering photometric aerosol detection.

Thermo Fisher 3600 TEOM PDM: The PDM was designed for use in US coal mines to give time-resolved measurements of coal miners exposure to respirable dust (Volkwein *et al.,* 2004). It is a miniaturized version of the PDM 1400 environmental monitor. A length of low particle loss sampling tube transports dust-laden air from the sample inlet located on the workers cap lamp to a respirable cyclone inlet on the main PDM unit worn on the waist. The sampled dust passes through a heater to remove excess moisture and is collected onto the filter of a TEOM mass sensor.

The mass change on the filter is calculated from the change in its frequency using first principles of physics (Patashnick *et al.,* 2002). The instrument is therefore not subject to uncertainties in measurement caused by changes in particle size, colour, or shape.

Modified PDM 3600: The PDM was modified to sample and measure larger sized particles in order to improve its inhalable dust measurement characteristics. This was achieved by a series of modifications: (i) removing the cyclone inlet from the PDM, (ii) removing the sensing unit from the main pump/display unit (and reconnecting using a length of flexible tubing and electrical cable), which allowed it to be positioned within the workers breathing zone and also allowed the sampled dust to be deposited directly onto the filter, and (iii) attaching a disc of electrostatic filter material to the top of the existing filter to improve the retention of larger particles. The effect of these modifications on the performance of the monitor was assessed by exposing the modified PDM to a range of sizes of aluminium oxide aerosol (13–58 μm mass median aerodynamic diameter MMAD Mark *et al*, 1985) generated inside a calm air chamber and comparing the response with a reference IOM gravimetric sampler located alongside.

The results of these modifications are shown in Fig. 1. With Modification 1, the PDM response clearly decreases compared with the reference IOM sampler as the particle size increases from 6 to 13 um MMAD. This is thought to be due to the larger particles being deposited within the PDM's somewhat convoluted internal aerosol transport tubes prior to reaching the sensor. With Modification 2, there is still an obvious decrease in PDM response with increasing particle size although not as marked as in the case of Modification 1. Visual inspection of the inside of the PDM sensor revealed that for larger particle sizes, there was a significant quantity of dust deposited around the filter stem that appeared to have been lost from the filter. It is generally observed that when large particles are collected onto filter medium, they are much

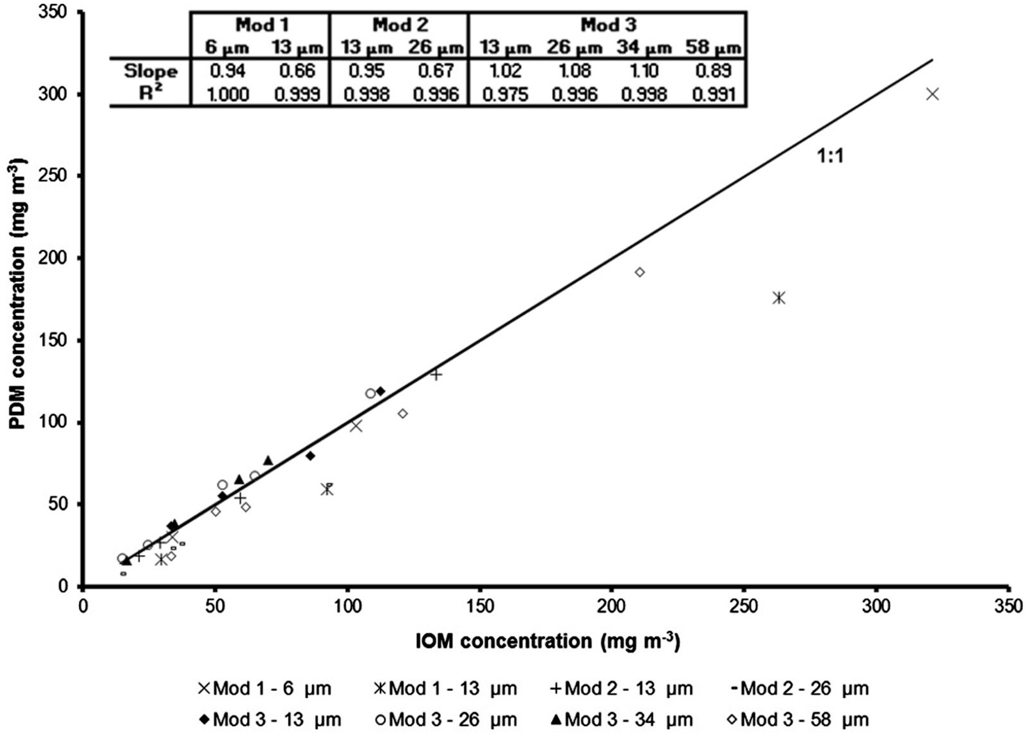

Fig. 1. Effects of modifications to PDM on instrument response.

more mobile than smaller particles, which are more likely to be trapped inside the matrix of the filter material or form a stable 'cake' of dust on the surface of the filter. The vibrating action of the filter is likely to exacerbate the effect and would explain the trend shown in Fig. 1. With Modification 3, there is a minimal decrease in PDM response with increasing particle size as observed with the original filter. There is also good linearity over the measured concentration ranges as shown by R^2 values close to unity. On inspection, there was no obvious loss of particles from the filter. During subsequent tests, modified filters were used with the modified personal PDM and also the TEOM model 1400 during all of the instrument comparison tests.

On the basis of these results, the modified PDM was used as a secondary standard in the laboratory and field for the real-time measurement of inhalable dust concentration.

There was concern that circumventing the internal drying tube and the positional sensors located inside the main unit might lead to operational problems in practice. However, if the concept of using the PDM as an inhalable monitor could be proven, then any such issues should not be insurmountable.

Static direct-reading monitors

TSI Dustrak DRX. The DRX simultaneously measures size-segregated mass fraction concentrations corresponding to PM1, PM2.5, PM10, respirable, thoracic, and total PM (up to 15 μm). It combines a light scattering photometer and an optical particle counter for mass fraction concentration measurements between 0.001 and 150 mg m^{-3} in the size range 0.1–15 μm. For the purpose of these tests, the flow rate was adjusted to 2 l min^{-1} so that, like the TSI Sidepak, an IOM adaptor could be connected to the inlet. The monitor is available in desktop and handheld versions, which have similar features. For the purpose of this work, the desktop model was used.

The DRX is calibrated using two CFs: a photometric CF (PCF) and a size CF (SCF). The PCF corrects for the photometric response difference between 'SAEJ726 Fine' standard test dust (for which the instrument is calibrated in the factory) and the aerosol being measured, whereas the SCF accounts for the aerodynamic size differences.

Thermo Fisher TEOM model 1400. This is a larger, freestanding version of the PDM 3600 that is predominantly used to monitor ambient air quality. Using a choice of inlets, it can measure PM10, PM2.5, PM1, or TSP concentrations. It has a measurement range of <5 μg m^{-3} to several g m^{-3} with a resolution of 0.1 μg m^{-3}. The real-time mass concentration averaging time can be set between 10 and 3600 s. The electrostatic filters used for the PDM 3600 were also used for the 1400 model.

Amherst aerodynamic particle inspector. The aerodynamic particle inspector (API) uses the time of flight principle to determine the size distribution of airborne dust and powders in real time (Niven, 1993) in the size range 0.20–700 μm. It measures the aerodynamic diameter of a particle. It should be noted that the high acceleration forces acting on particles entering the inlet can cause them to deagglomerate thereby making them appear smaller than when airborne.

Laboratory tests

Test dusts. Dusts representing an inhalable health hazard in the workplace were used in the tests, which included: barley grain dust, softwood and hardwood dusts, flour dust, and aluminium oxide powder (Mark *et al*, 1985). Barley grain was milled to produce grain flour using a portable grain mill (Walder Biotech Ltd). This was then sieved to produce three distinct ranges of particle sizes (<75, 75–106, 106–180 μm). Plain white wheat flour was purchased from a supermarket and was also sieved to produce the same range of particle sizes. Three size fractions of 'Fibretron' softwood dust (60, 120, 180) were obtained from WTL International Ltd, with particle sizes of <150, <125, and <90 μm, respectively. This is essentially a mixture of spruce and pine softwood dust produced by grinding and sifting. Two hardwoods were also chosen for testing. The first was afrormosia, which was already available in different size fractions, produced during sanding using 80, 100, and 120 grades of sandpaper. The second was beech wood dust and was produced by sanding a plank of beech using a belt sander fitted with 40 and 80 grade sanding belts. The dust produced was captured using a vacuum cleaner attached to the dust extraction port on the side of the sander. The wood dusts were not sieved to classify them further.

Experimental methodology. Tests were carried out inside a large dust tunnel so that the dust monitors and samplers were exposed to dust concentrations under controlled steady-state conditions. The samplers and monitors were mounted onto a 'Little Anne' manikin (Laerdal Medical AS) placed inside the tunnel on a rotating turntable.

'Little Anne' manikin: The 'Little Anne' manikin is normally used as an aid to provide adult cardiopulmonary resuscitation training to medical staff. It comprises the upper torso and was therefore easily accommodated into the dust tunnel giving a blockage of ~13% of the tunnel's total cross-sectional area. This is less than the maximum tunnel blockage of 15% quoted in EN13205 (CEN, 2002). The manikin was attached to the rotating turntable via a length of bar, which was placed beneath the tunnel to minimize blockage. The turntable and manikin rotated at 1 r.p.m. with a reciprocating action so that any sampling biases caused by the relative motion of dust samplers to air flow were minimized. The samplers and dust monitors were mounted on the front and rear of the manikin with hook and loop tape for ease of attachment and release.

Large dust tunnel: The large dust tunnel can be configured to create working sections with different dimensions. For the purpose of these tests, a cross-section of 1.5 × 1.0 m was used. Air is blown down the tunnel using a large centrifugal fan to a bank of high efficiency particulate air HEPA filters at the other end that remove any airborne particles, so that clean air is recirculated. Metal mixing grids immediately downstream of the fan help to create a uniform flow through the tunnel. In order to compare sampler/dust monitor performance, it is important that air velocity and dust concentration within the sampling region are constant and uniform.

The tunnel velocity profile was determined using a calibrated hot wire anemometer (Velocicalc, TSI Inc.) situated at the manikin position. The anemometer probe was initially placed 15 cm from the side and 10 cm from the top of the tunnel and was then traversed every 30 cm horizontally and 20 cm vertically to give a total of 25 measurements. A calibrated reference anemometer (Velocicalc, TSI Inc.) was also placed centrally inside the tunnel just downstream of the traversing probe to correct for any temporal changes in velocity. At a nominal velocity of 0.5 m s^{-1}, the profile was uniform over the tunnel cross-section with a coefficient of variation (CV) of 7% from a mean value of 0.57 m s^{-1}. The profile was slightly better over the area that the manikin occupied with a CV of 6% from a mean value of 0.56 m s^{-1}.

Dust was injected as close as possible to the manikin to minimize losses caused by gravitational settling while maintaining a uniform dust concentration profile over the sampling area. Aloxite dust (MMAD = 58 µm) was metered using a screw feed system (March Systems Ltd) into a venturi dust pump powered by compressed air, which dispersed and injected the dust into the tunnel via a length of flexible tubing. This gave a relatively constant dust concentration inside the tunnel with time. The concentration of dust inside the tunnel could be adjusted by changing the size or rotational speed of the screw.

Six IOM gravimetric samplers were placed on the upper torso of the manikin (three on the front and three on the back) within the breathing zone. The resultant variation in concentration was 6–8%, which was regarded as acceptable because it was assumed that the variation would be as good if not better for smaller particle sizes that would be generated during the dust monitor comparison tests.

Tunnel experiments: The dust monitors and reference gravimetric samplers were arranged either side of the manikin as shown in Fig. 2 at the position inside the tunnel shown in Fig. 3. The TEOM model 1400 monitor was positioned upstream of the manikin with the main measurement unit located below the tunnel and the heated column passing upwards through the floor. The TEOM column was fitted with a sharp-edged isokinetic inlet probe (Willeke and Baron, 2001) that faced into the tunnel so that at its operating flow rate of 3 l min^{-1}, the velocity into the probe matched that of the air velocity down the tunnel (0.5 m s^{-1}). Alongside the TEOM was placed a reference isokinetic gravimetric probe, the Microdust Pro probe and a reference SIMPEDS personal cyclone sampler. The API was also placed below the tunnel with its inlet connected to a length of copper pipe passing vertically through the tunnel floor and positioned alongside the manikin. Attached to this was another isokinetic probe.

The DRX dust monitor was placed on the tunnel floor, close to the manikin and was connected to an IOM inlet located on the manikin using a length of low particle loss tubing (see Fig. 2). The tube length was minimized to reduce particle losses while allowing for the rotational movement of the manikin.

The IOM samplers, cyclone samplers, isokinetic sampler, and Respicon were loaded with clean glass fibre filters, which were conditioned and weighed before and after exposure. Air was drawn through these at flow rates of 2, 2.2, 2.36, and 3.11 l min^{-1}, respectively, using flow regulated personal sampling pumps. The flow was checked before and after each test using a calibrated bubble flow meter

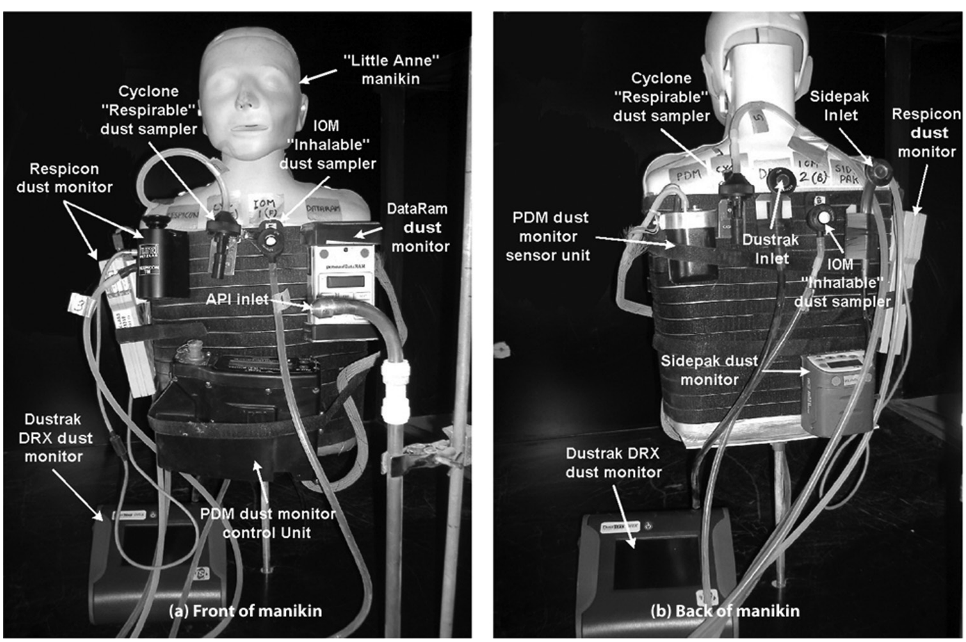

Fig. 2. Position of dust monitors and samplers on the 'Little Anne' manikin.

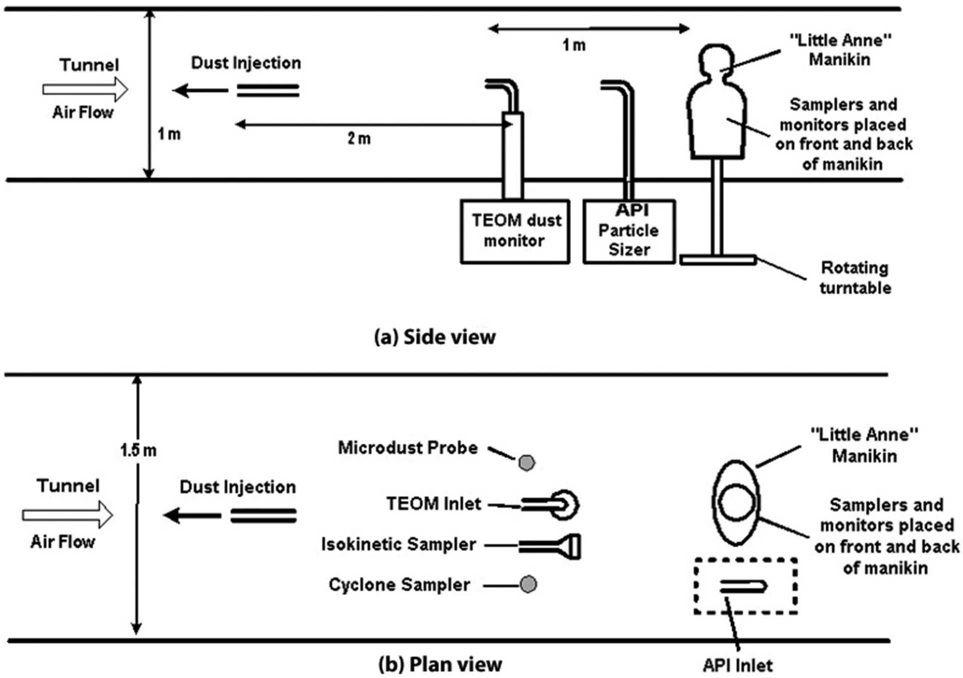

Fig. 3. Schematic of the dust tunnel showing the relative positions of the dust injector, dust monitors, and reference samplers.

(Gilian Ltd). Each photometer-type dust monitor was zeroed prior to testing and programmed to record a measurement every 1–5 s. The PDM was programmed to make a measurement every minute, which was regarded as the minimum time required to collect a detectable amount of dust onto the PDM filter, at the lowest concentrations generated inside the tunnel. The TEOM 1400 was set to average and log every 15 s, which was possible because it is ~10 times more sensitive than the PDM as detailed in the manufacturer's product specification sheets. The PDM and TEOM 1400 were fitted with new 'modified' filters prior to each test. Flows into the TSI Sidepak and DRX dust monitors were adjusted to 2 l min^{-1} so that the attached IOM inlets operated at the correct flow rate. The Microdust Pro dust monitor was calibrated before each test using the supplied calibration element.

The tunnel velocity was adjusted to 0.5 m s^{-1} before each test and the smallest particle size of each test dust was continuously fed into the tunnel using the screw feed/injector system, which resulted in a relatively constant dust concentration inside the tunnel. In order to investigate the effects of sudden changes in particle size on instrument calibration, quantities of larger particles were periodically injected into the tunnel to produce concentration peaks. This was achieved by measuring between 1 and 9 g of dust into small glass beakers (the amount depending on the particle size and density of the dust being investigated). For each particle size, one beaker of dust was introduced every 15 s for a period of 3 min to even out any positional effects caused by the rotation of the manikin. This procedure was repeated for each additional peak concentration. Each test lasted for ~60 min, long enough for a weighable dust sample to be obtained by the inhalable samplers. The API was used to measure the size of the airborne dust throughout each test.

Workplace measurements

The next stage of the work was to test a selection of dust monitors at three workplaces:

- A small wholesale family-run bakery business employing around 15 workers that manufactures a variety of confectionery products including bread, cakes, pork pies, and sausage rolls. Although automated machinery such as ingredient mixers, dough cutters/rollers, and flour sifters were used, much of the production was carried out by hand.

- A small woodworking workshop (joinery) employing around 10 workers, which produced mainly office furniture using a variety of wood types and composites. Fixed machinery and handheld power tools were used and some manual work (sanding, planing etc.) was also carried out.
- A large grain mill where organically produced grain was milled, compounded, made into pellets, and bagged to produce feed primarily for farm livestock.

The types of dust produced inside these workplaces were largely consistent with those generated during the laboratory trials. Wherever possible, the site visits comprised a pre-visit and a main visit to each workplace. The aim of the pre-visit was to observe the work activities/processes being carried out and to identify sources of dust emissions likely to produce high concentrations of airborne dust. This enabled the best sampling strategy to be determined for the main visit. A limited amount of static dust monitoring was also carried out during the pre-visits. Personal exposure and static measurements were usually carried out during the main visit; video exposure monitoring (Rosén *et al.,* 2005) was also performed at the bakery and joinery.

The PDM and Respicon TM monitors were chosen for the site visits because they showed good agreement with the IOM sampler for a range of dust types during the laboratory trials (see Results section). The DataRam was also used to investigate errors that occur when photometer-type dust monitors are used to estimate airborne inhalable dust concentrations. IOM gravimetric samplers were used to give time-weighted average reference measurements of inhalable dust concentration.

Methodology. The PDM (or Respicon TM), DataRam, and IOM sampler were attached to a worker within the breathing zone using a body harness to measure personal exposure. Additional static measurements of airborne dust were also made nearby using the PDM (or Respicon TM), DataRam, and IOM sampler. The measurements lasted for long enough so that: (i) the dust monitors could identify and measure airborne dust concentrations produced by different work activities and (ii) the IOM samplers collected a weighable quantity of dust. The instruments were operated in the same way as in the laboratory trials. Any variation in the CF for the DataRam dust monitor was determined throughout each site visit by using the PDM or the Respicon TM as a secondary reference measurement of inhalable dust concentration. This has been shown to be valid for the PDM (see Fig. 1) and the Respicon TM (see Fig. 5), where the temporal variation measured by the PDM and Respicon TM are reasonably close for a wide range of particle sizes.

RESULTS AND DISCUSSION

Laboratory measurements

Average monitor response versus reference gravimetric concentration. The average concentration measured inside the dust tunnel over the duration of a test by each monitor for each type of dust was plotted against the reference IOM inhalable concentration and these are shown in Fig. 4 for the PDM, Respicon, TEOM 1400, and DataRam. The Microdust Pro, DRX, and Sidepak results are not shown because it would make the figure too complicated because the degree of scatter in results was similar to the DataRam in each instance. The results are also summarized in Table 1, which shows the monitor response, calculated from the slopes of the graphs, the intercept, and the coefficient of determination, R^2 derived from a linear regression of the data.

The PDM and Respicon (inhalable) monitors showed good linearity when compared with the reference IOM sampler, indicated by R^2 values of 0.965 and 0.976, respectively. This is despite the data consisting of a variety of dust types and a wide spread of particle sizes, as shown in Table 2. It indicates that both these instruments are relatively unaffected by changes in the physical properties of the dust. The PDM R^2 value is comparable and slightly closer to 1 than those observed by Volkwein et al. (2004) during their initial laboratory and field assessment of the prototype PDM. On average, the PDM underestimated the inhalable concentration by ~27% compared with the IOM sampler, which could be caused by the aspiration characteristics of the PDM inlet or possibly some dust was lost from the surface of the filter, despite the modifications made. The Respicon inhalable agreed within ~6% of the reference IOM concentration. The TEOM 1400 was in closer agreement with the reference IOM sampler than the PDM, possibly because the TEOM was positioned ~1 m upstream of the PDM where the concentration in the tunnel was slightly higher. There was more scatter in the TEOM results indicated by a lower R^2 value, which could be due to

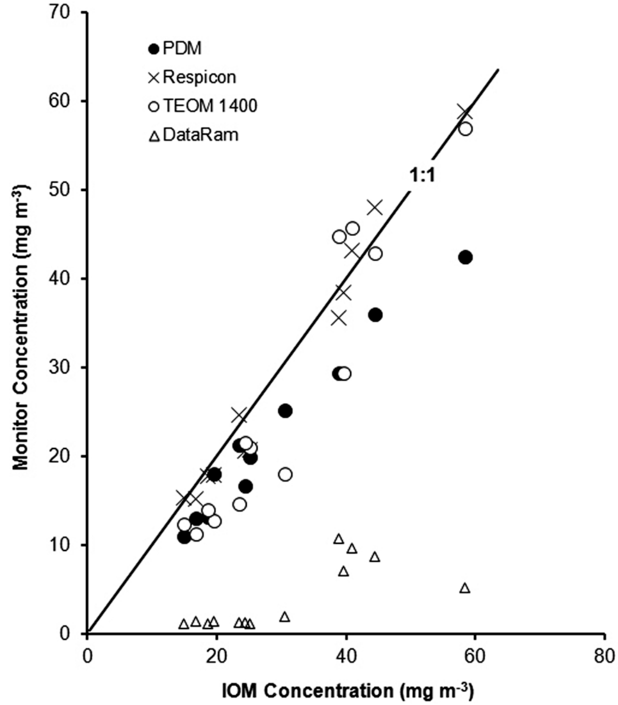

Fig. 4. Variation in average monitor concentration with reference IOM concentration for all dust types.

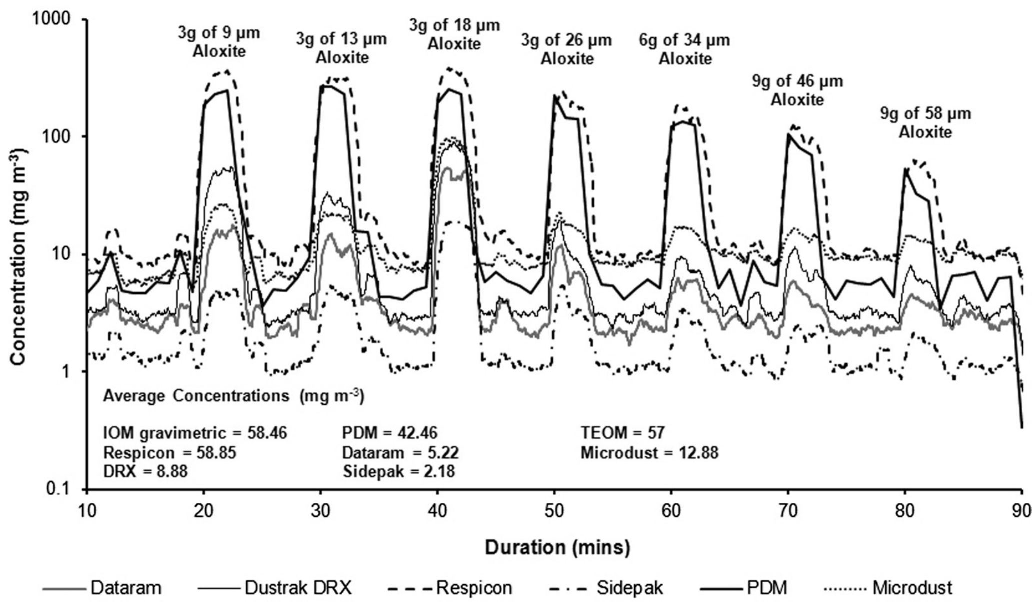

Fig. 5. Temporal variation in monitor response with time for aerosols of Aloxite.

particle losses in the TEOM's isokinetic inlet and/or connecting bend, which will vary as the particle size changes. Also, the difference in dust concentration measured at the TEOM and IOM sampling positions changes with particle size due to gravitational settling and may contribute to the observed scatter. There was little correlation between the TEOM and gravimetric isokinetic sampler and so

Table 1. Response of dust monitors to the inhalable fraction when challenged with all dusts.

Dust monitor	Monitor response (monitor/IOM)	Coefficient of determination (R^2)	Intercept through y-axis
PDM	0.73	0.965	1.41
TEOM 1400	1.14	0.899	−8.16
Respicon	1.06	0.976	−2.67
DataRam	0.21	0.554	−2.48
Sidepak	0.1	0.548	−1.45
Microdust	0.429	0.494	−2.5
Dustrak DRX	0.22	0.894	−2.78

Table 2. Variation in CF throughout laboratory measurements.

Dust event	Dust type (nominal particle size)	Aerodynamic particle size (µm)		CFs (PDM response/monitor response)					
		Mass median	Number median	DataRam	DRX	Micro dust	Sidepak	Respicon	TEOM
Background	Aloxite (6 µm)	5.02	1.2	3.80	2.53	1.38	7.06	1.27	1.35
Peak 1	Aloxite (6 µm + 9 µm)	12.52	3.72	14.29	4.60	9.64	48.42	0.68	0.75
Peak 2	Aloxite (6 µm + 13 µm)	16.18	3.57	18.22	8.39	12.23	51.39	0.83	0.75
Peak 3	Aloxite (6 µm + 18 µm)	18.8	1.61	4.69	2.86	1.58	13.40	0.66	0.67
Peak 4	Aloxite (6 µm + 26 µm)	22.39	2.14	19.72	13.02	9.95	43.58	0.92	0.81
Peak 5	Aloxite (6 µm + 34 µm)	28.51	2.1	22.55	18.56	8.07	43.80	0.72	0.62
Peak 6	Aloxite (6 µm + 46 µm)	37.89	1.94	25.57	11.23	6.93	44.34	0.88	0.88
Peak 7	Aloxite (6 µm + 58 µm)	43.37	1.76	13.08	8.34	3.93	29.70	0.95	0.87
			Average	15.24	8.69	6.71	35.21	0.86	0.84
			CV (%)	51.84	63.24	60.08	47.55	22.86	26.96
Background	Barley (<75 µm)	25.66	8.27	14.55	9.2	4.82	83.88	0.82	1.29
Peak 1	Barley (<75 µm + 75–106 µm)	27.93	29.84	14.23	10.95	6.14	66.14	0.81	1.41
Peak 2	Barley (<75 µm + 106–180 µm)	41.95	28.78	15.75	9.82	6.22	83.07	0.86	1.57
Peak 3	Barley (<75 µm + 75–106 µm)	46.83	30.41	18.79	15.08	7.74	127.08	1.02	1.59
Peak 4	Barley (<75 µm + 106–180 µm)	97.08	28.6	12.58	9.31	5.45	65.83	0.74	1.32
			Average	15.18	10.87	6.07	85.20	0.85	1.44
			CV (%)	15.24	22.56	17.97	29.34	12.28	9.67
Background	Flour 0–75 um	28.18	7.83	28.27	14.69	3.73	471.2	1.13	0.95
Peak 1	Flour 0–75 um (75–106 um)	29.57	9.03	17.05	13.02	3.94	169.9	0.88	0.86
Peak 2	Flour 0–75 um (106–180 um)	30.4	9.64	15.52	16.51	4.74	145.7	0.96	0.99
Peak 3	Flour 0–75 um (75–106 um)	28.15	9.53	17.06	16.46	5.07	133.2	1.01	1.02
Peak 4	Flour 0–75 um (106–180 um)	28.41	9.73	14.64	16.35	5.27	118.8	0.94	1.06
			Average	18.51	15.41	4.55	207.76	0.98	0.98
			CV (%)	30.01	9.96	15.02	71.46	9.55	7.82
Background	Fibretron (180)	22.61	6.73	12.91	9.96	2.55	30.3	0.64	0.78
Peak 1	Fibretron (180 + 120)	26.66	7.91	12.89	14.54	3.25	32.34	0.81	1
Peak 2	Fibretron (180 + 60)	31.74	5.94	7.67	8.21	2.3	20.8	0.7	0.95
Peak 3	Fibretron (180 + 120)	71.68	8.16	12.82	17.43	4.48	36.56	0.82	1.13
Peak 4	Fibretron (180 + 60)	23.63	5.76	7.27	7.46	2.19	21.57	0.62	0.94
			Average	10.71	11.52	2.95	28.31	0.72	0.96
			CV (%)	27.66	37.32	32.07	24.35	13.01	13.11
Background	Afrormosia (120 grade sandpaper)	14.66	5.35	9.37	7.89	2.75		0.73	0.8

(Continued)

Table 2. *Continued*

Dust event	Dust type (nominal particle size)	Aerodynamic particle size (µm)		CFs (PDM response/monitor response)					
		Mass median	Number median	DataRam	DRX	Micro dust	Sidepak	Respicon	TEOM
Peak 1	Afrormosia (120 + 100 grade sandpaper)	21.59	5.01	8.66	9.87	3.85		0.63	0.87
Peak 2	Afrormosia (120 + 80 grade sandpaper)	19.91	4.93	7.59	7.65	2.98		0.59	0.77
Peak 3	Afrormosia (120 + 80 grade beech)	19.68	5.4	8.87	9.51	3.62		0.6	0.84
Peak 4	Afrormosia (120 + 40 grade beech)	18.67	5.12	9.13	8.79	3.83		0.7	0.96
Peak 5	Afrormosia (120 + fibretron 120)	17.13	4.78	8.2	10.11	3.55		0.78	0.99
			Average	8.64	8.97	3.43		0.67	0.87
			CV (%)	7.54	11.52	13.37		11.43	10.04

these are not reported. The reasons for this are not known, however, the TEOM agreed quite well with the IOM sampler results and it is possible, therefore, that the errors are with the isokinetic sampler.

There was considerable scatter in the measurements made with the photometer-type dust monitors compared with the reference IOM sampler with no obvious correlation between the two, as indicated by low R^2 values when a linear fit was applied to the data. This is expected because photometer-type dust monitors respond essentially to the respirable fraction and progressively underestimate the inhalable concentration as the particle size increases beyond the respirable size range (Thorpe, 2007). Table 2 shows that in most instances, the majority of airborne dust was above the respirable size range.

For most tests, the fraction of respirable dust compared with inhalable airborne dust was very low, which meant that the cyclone samplers often did not collect enough dust to allow accurate gravimetric measurements to be made and so these results are not reported here.

Temporal variations in dust monitor concentration. Figure 5 shows the variation in monitor concentration with time for Aloxite dust. Graphs for the other dust types are not presented here, but show similar trends. Peaks are clearly visible when different sizes of dust were injected. Each photometer-type dust monitor significantly underestimated the inhalable dust concentration as measured by the PDM, and Respicon (inhalable) dust monitors, which was expected because most of the aerosols contained particles larger than respirable, which would not be detected. The plot of TEOM concentration is not shown on Fig. 5 to prevent confusion, but was similar to the Respicon inhalable and PDM.

The DRX dust monitor underestimated the inhalable concentration although by not as much as the DataRam and Sidepak (the Microdust read higher because it is factory calibrated against the larger TSP fraction). This is because the DRX only accurately detects particles up to 15 µm in size and so does not measure the fraction of particles in the inhalable size range that are above this. Also, it requires calibration when used in different dust atmospheres and is, therefore, subject to the limitations found with other photometer-type monitors when the dust size or type changes.

Changes in dust monitor CF. The variation in photometer CF was determined throughout each test by using the PDM as the reference measurement of inhalable concentration (although the Respicon could also have been used). CF is defined as the ratio of PDM concentration to photometer concentration. This assumes that the PDM's performance is unaffected by dust type and particle size, which is largely confirmed by the results shown in Fig. 1. Table 2 summarizes the average CFs determined for each dust type and also identifies changes in CF caused by the introduction of different particle sizes described as 'dust events'. It also shows the corresponding particle size measured using the API.

For Aloxite dust, the CF for each photometer-type dust monitor varied significantly throughout a test depending on the size of aerosol produced during any particular event. For example, for the DataRam dust monitor, the CF changed from 3.8, when measuring Aloxite particles with a nominal MMAD of 6 µm (MMAD = 5 µm measured using the API), to 25.5 for a mixture of 6 and 46 µm Aloxite particles (MMAD = 37.9 µm measured using the API). The average CF throughout the test was 8.3. A change in CF was also observed for Fibretron wood dust, although there were no significant changes observed for the other dusts.

These effects of particle size on CF of photometers are further illustrated in Fig. 6, which shows that the concentration indications of a photometer are unreliable if particle size is varying substantially. The figure shows the PDM output (considered here as the 'true' inhalable measurement), the DataRam output (as an example of a photometer), and the DataRam output corrected by a CF calculated from the average gravimetric concentration during the test. Outside the pulses of coarse dust, with the monitors exposed to 6 um Aloxite, the DataRam CF record is above the PDM's, but for seven of the eight injections of coarser dust, the DataRam CF record is below the PDM's. This is because the DataRam, like other photometers, is much more sensitive to the 6 um dust than to the coarser grades, but the PDM's output is relatively independent of particle size (see Fig. 1). The third peak is an exception. This seems to be because the nominally 18 um Aloxite contains a substantial amount of finer dust, confirmed by the relatively low median sizes for this dust in Table 2. Although not shown here, a similar but less pronounced trend was also observed for Fibretron wood dust, whereas for flour, barley grain dust, and mixed wood dusts, the calibrated DataRam concentrations followed the PDM concentration quite closely. Similar trends were observed for all of the other dust monitors and are summarized in Table 2. It is clear that the application of an average CF to photometer-type dust monitors to determine variations in inhalable airborne concentration can result in significant and unknown errors if the size of the aerosol changes.

Workplace measurements

Joinery. During the pre-visit, various machining operations were being carried out including sawing, routing, drilling, and sanding. The DataRam and PDM were placed on a pedestal drill stand, located close to groove cutting and guillotining operations. The groove cutter was fitted with local exhaust ventilation (LEV) provided by a bag-type dust extractor into which the captured dust was collected. On inspection of the dust collection bag, it was observed that there was a large hole ~3 cm in diameter through which the collected dust escaped and was released into the workshop, which explains the large Peak C shown in Fig. 7. Table 3 shows that the DataRam CF increased during the machining operations (Peaks A, B, and C) compared with that obtained from measurements of the 'background' dust concentration when little or no work was being carried out. Figure 7 shows the effect of applying an average CF to the DataRam measurements: the background inhalable concentrations are overestimated and the inhalable peak concentrations are underestimated.

For the main visit, a PDM, DataRam, and IOM sampler were attached to a worker using a body harness. His main activity was the construction of a display stand made from medium density fibreboard (MDF) during which he used a variety of handheld power tools. In addition, the Respicon, another DataRam, and another IOM sampler

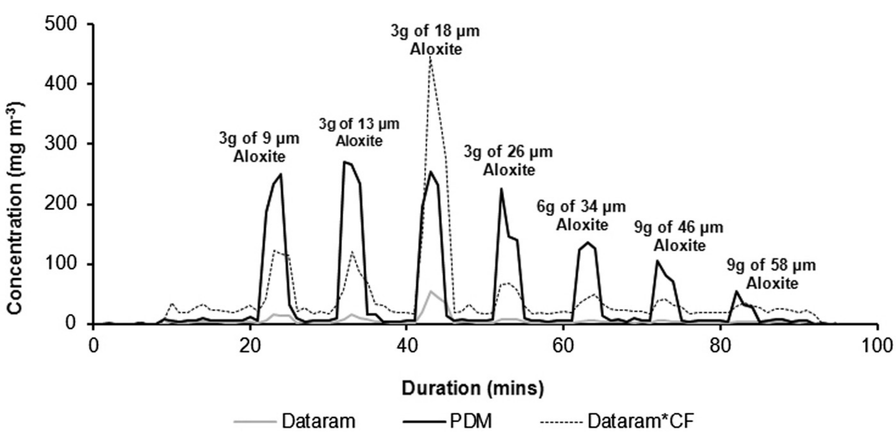

Fig. 6. Temporal variation of DataRam measurements with and without average CF applied and PDM monitor for Aloxite dust.

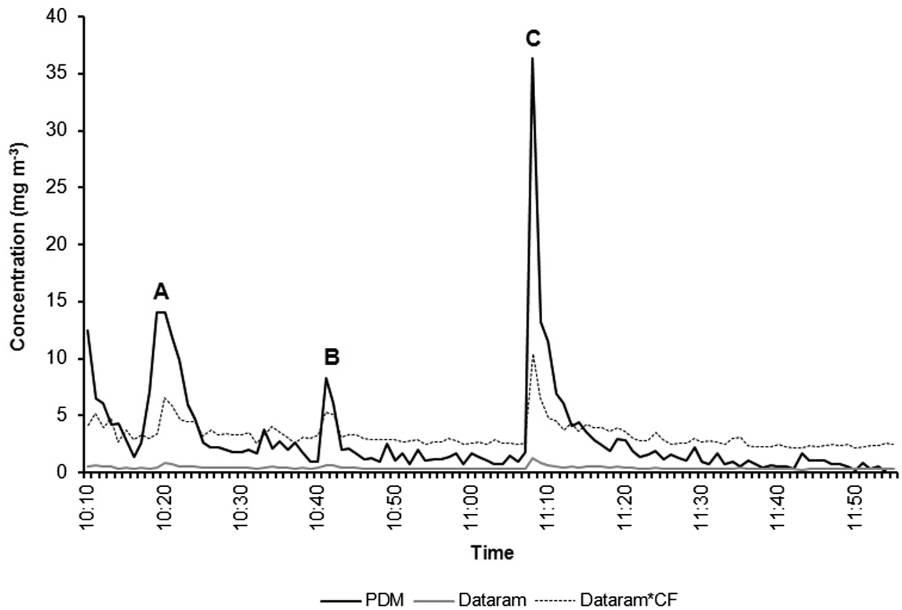

Fig. 7. Application of an average CF to DataRam measurements to determine temporal variations in inhalable concentration—joinery pre-visit, static sampling (see Table 4 for description of activities A–C).

were located at static positions around the workshop. During the measurements, it was observed that the PDM was sensitive to movement, resulting in some erroneous readings. For this reason, the PDM results are not presented or discussed here and it was not used for personal dust exposure measurements during subsequent site visits. Table 3 describes the concentration peaks measured at the various static sampling locations around the workshop, and clearly shows that the DataRam CF varied depending on the work activity.

Bakery. During the pre-visit, the production of pita bread was the main work activity. Here, pitta bread dough is cut to size, rolled into shape, and flour is sprinkled onto the surface using a sifting machine and by hand. Excess flour is then manually brushed off the surface and the resultant dust cloud is directed towards an LEV extractor hood in an attempt to keep airborne dust levels to a minimum. Two DataRams, the Respicon TM, the PDM, and two IOM reference samplers were located downstream of the hood, which was lifted briefly during the measurements. After approximately an hour, the PDM, one DataRam, and one IOM sampler were moved close to the flour-sifting machine.

Concentration peaks and DataRam CFs are summarized in Table 4. The CF was determined using both the PDM and the Respicon as the reference measurement of inhalable concentration. It can be seen that the CF varied appreciably depending on the work activity being carried out. Of particular interest is peak E where the CF is considerably lower than all other work activities. This indicates that the airborne dust consisted mostly of small particles and it is thought that this peak might have been produced by diesel exhaust fume because the bakery's delivery vans were being loaded up close by at about this time. As an aside, Table 4 also shows that according to the PDM and Respicon, the extractor hood removed ~70% of the inhalable airborne dust.

During the main visit, the production of pitta bread was once again the principle source of airborne flour dust. However, during this visit, personal exposure rather than static dust measurements were made. The Respicon TM, DataRam, and IOM sampler were attached to a worker using the body harness, who was making the pitta bread dough. Table 4 shows once again that DataRam CF varied depending on work activity. The application of an average CF to the DataRam measurements resulted in an under or overestimation of airborne inhalable dust compared with the Respicon inhalable depending on the work activity, as shown in Fig. 8. For example, peak A was hardly detected by the DataRam, whereas the background inhalable

Table 3. Estimated task-averaged DataRam CFs measured during static sampling at a joinery.

Work activity	Concentration ($mg\ m^{-3}$)			CF
	PDM	Respicon (inhalable)	DataRam	
Visit 1 (pre-visit)				
Background	1.29		0.33	3.87
A—Grooves cut into wooden lattes (three bag extractor switched on)	7.42		0.51	14.61
B—Guillotining mitres on wood	4.33		0.51	8.53
C—Grooves cut into wooden lattes (three bag extractor switched on)	13.93		0.70	19.88
Average	3.25		0.39	8.25
Visit 2 (main visit)				
Background		0.56	0.04	12.69
A—Instruments close to a circular mitre saw				
Peak 1		4.81	0.20	23.50
Peak 2		3.59	0.09	39.94
Peak 3		2.61	0.17	14.95
B—Instruments close to a large circular saw				
Peak 4		3.98	0.14	27.81
Peak 5		3.47	0.08	42.86
C—Instruments close to a worker carrying out various hand machining operations				
Peak 6		4.58	0.11	40.61
Peak 7		3.76	0.10	36.31
Peak 8		4.52	0.23	19.63
Peak 9		2.75	0.15	18.55
Peak 10		1.41	0.14	9.79
D—Instruments close to a circular mitre saw				
Peak 11		2.32	0.11	20.45
Peak 12		6.29	0.15	42.38
Peak 13		1.43	0.11	13.12
Peak 14		1.29	0.09	13.64
Peak 15		1.54	0.09	17.15
E—Instruments close to a worker carrying out various hand machining operations				
Peak 16		1.88	0.07	28.15
Peak 17		4.32	0.10	42.23
Peak 18		6.67	0.40	16.86
Peak 19		8.02	0.70	11.48
Peak 20		3.39	0.11	31.31
Peak 21		5.03	0.16	31.22
Average over total test duration		2.78	0.25	11.32

concentration was overestimated. The DataRam also underestimated the concentration peaks 1–10, measured during pitta bread production but overestimated peak C.

Grain mill. During the visit, a major cleaning exercise was underway using compressed air lines to remove dust that had accumulated on the outside of the grain hoppers. An industrial vacuum cleaner was used to capture the dust at source, which was not very effective. The resulting concentration of airborne dust inside the mill was very high comprising mainly large particles that were clearly visible in the air. The PDM, DataRam, and IOM sampler were located on a gantry next to a grain hopper below where the cleaning was taking place. The Respicon, another DataRam, and another IOM sampler were placed close by on the

Table 4. Estimated task-averaged DataRam CFs measured during static sampling (pre-visit) and personal sampling (main visit) at a bakery.

Work activity	Concentration (mg m^{-3})				CF	
	Respicon inhalable	DataRam 1	PDM inhalable	DataRam 2	Respicon inhalable	PDM
Visit 1—Pre-visit (static sampling)						
Background	0.98	0.083	2.20	0.066	11.88	33.08
A—Brushed excess flour from surface of pita bread extractor on	2.94	0.103	4.53	0.109	28.48	41.52
B—Brushed excess flour from surface of pita bread extractor off	12.15	0.589	14.03	0.484	20.63	29.02
C—Ingredients mixed using a mixing machine. Floor swept close by	8.31	0.358	13.80	0.347	23.23	39.75
D—Dough moved from mixing machine to warmer/fermenter	5.59	0.289	15.99	0.366	19.33	43.73
E—Dough cut and weighed	1.20	0.202	2.63	0.188	5.94	14.03
F—PDM, DataRam, and IOM sampler moved to flour-sifting machine			9.92	0.214		46.45
G—Flour sifter brushed clean	4.53	0.186	12.59	0.200	24.33	62.79
H—Floor swept	1.32	0.107			12.33	
Average over total test duration	2.26	0.136	4.72	0.137	16.60	34.44
Visit 2—Main visit (personal sampling)						
Background	0.41	0.508			0.80	
A—Unknown source	0.81	0.473			1.72	
B—Peaks 1—10 caused by worker sprinkling flour over pita dough						
Peak 1	3.24	0.165			19.60	
Peak 2	2.65	0.165			16.01	
Peak 3	2.90	0.308			9.43	
Peak 4	2.62	0.217			12.08	
Peak 5	2.52	0.234			10.78	
Peak 6	3.10	0.245			12.64	
Peak 7	3.19	0.306			10.44	
Peak 8	3.17	0.256			12.40	
Peak 9	2.17	0.197			11.01	
Peak 10	2.40	0.224			10.71	
C—Worker filled sifter hopper	2.04	0.810			2.52	
D—Worker placed down tray containing flour creating a cloud of dust	7.27	0.616			11.80	
Average over total test duration	1.45	0.433			3.34	

top of steps leading to the gantry. An increase in the CF was observed when the cleaning process was underway, as shown in Table 5. This was much more evident for the PDM than the Respicon, almost certainly because the PDM oversampled large particles that entered the inlet due to gravitational settling. A suitably designed 'inhalable' inlet is required to eliminate this problem. Figure 9 shows a smaller difference between the calibrated DataRam and Respicon inhalable measurements than were observed during other site visits, probably because the size of the airborne dust changed little throughout the measurements. Although not shown here, similar trends to those shown in Fig. 9 were observed for the modified PDM.

Comparison of average PDM, Respicon inhalable, DataRam, and IOM concentration measurements. Table 6 shows the average concentration measurements made with the PDM, Respicon inhalable, and DataRam dust monitors, and the IOM sampler, during the site visits. Not surprisingly, the DataRam measurement was always much lower than that of the IOM sampler. The response of the PDM was variable when compared with the reference IOM sampler. It was

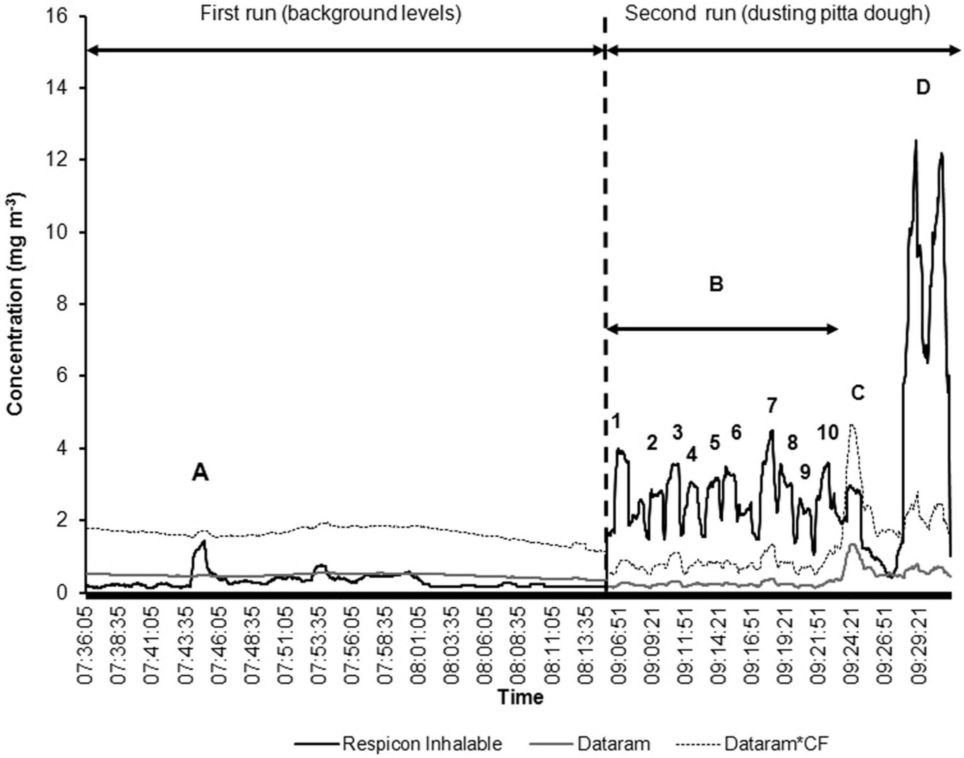

Fig. 8. Application of an average CF to DataRam measurements to determine temporal variations in inhalable concentration—bakery main sampling exercise, personal sampling (see Table 5 for description of activities A–D).

Table 5. Estimated task-averaged DataRam CFs measured during static sampling at a grain mill.

Work activity	Concentration (mg m^{-3})				CF	
	PDM inhalable	DataRam 1	Respicon inhalable	DataRam 2	PDM	Respicon
Background	2.46	0.079	1.02	0.044	31.2	22.8
A—Cloud of dust generated on top of gantry by worker movement	3.85	0.151			25.5	
B—Outside of hopper cleaned using compressed air line	49.62	0.412	29.25	0.645	120.4	45.3
C—Outside of hopper cleaned using compressed air line	172.47	1.811	145.97	4.436	95.2	32.9
Average over total test duration	28.66	0.375	20.43	0.620	76.4	33.0

considerably lower than the IOM sampler during the main visit to the joinery, owing to negative readings caused by the PDMs sensitivity to movement during personal sampling. During the pre-visit to the bakery, it read ~25% lower than the IOM sampler, which is consistent with the laboratory measurements. At the grain mill, it was considerably higher than the IOM sampler, likely due to oversampling caused by particles entering the inlet by gravimetric settling. Measurements of inhalable concentration using the Respicon were consistently close to the IOM sampler during the site visits, which is also consistent with the laboratory measurements.

SUMMARY AND CONCLUSIONS

Laboratory measurements

Photometer-type direct-reading dust monitors consistently underestimated the measurement of inhalable airborne dust concentration. This was

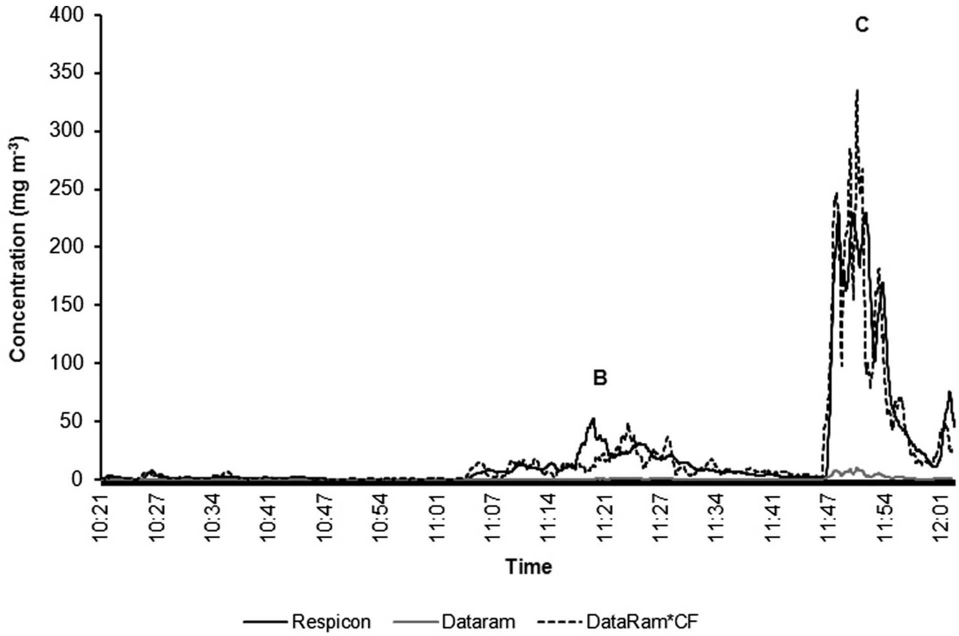

Fig. 9. Application of an average CF to DataRam measurements to determine temporal variations in inhalable concentration—grain mill, static sampling (see Table 6 for description of activities B and C).

Table 6. Average concentrations measured during site visits.

Work site	Average concentrations (mg m^{-3})								Inhalable monitor/ IOM	
	PDM	DataRam	IOM	Respicon			DataRam	IOM	PDM	Respicon inhalable
				Respirable	Thoracic	Inhalable				
Joinery—Main visit	3.63 (P)	0.35 (P)	6.89 (P)	0.57 (S)	1.25 (S)	2.27 (S)	0.24 (S)	1.87 (S)	0.53	1.22
Bakery—Pre-visit	2.64 (S)	0.14 (S)	3.57 (S)	0.1 (S)	0.31 (S)	2.26 (S)	0.14 (S)	2.32 (S)	0.74	0.97
Animal feed plant	23.52 (S)	0.37 (S)	14.55 (S)	0.53 (S)	3.33 (S)	20.43 (S)	0.62 (S)	23.22 (S)	1.61	0.88
							Averages		0.96	1.02
							Standard deviation		0.57	0.18
							CV		59.6	17.2

(P) = personal samplers/monitors; (S) = static samplers/monitors.

not surprising because they are not capable of detecting particles much beyond the respirable size range and indeed most manufacturers do not claim accuracy for measurement of inhalable dust concentration. It has been shown that scaling measurements made with photometer-type dust monitors, by applying average CFs determined throughout the measurement period using reference inhalable gravimetric samplers, can lead to appreciable real-time measurement errors if the particle size changes significantly throughout the measurement. Conversely, if the aerosol properties do not change during a measurement, then the use of an average CF would probably be acceptable.

A personal TEOM (PDM 3600) designed for use in US coal mines to measure worker exposure to respirable coal dust has been modified for use as an inhalable dust monitor. During laboratory trials, the modified PDM showed good linearity and estimated the inhalable airborne dust concentration to 73% of a reference IOM gravimetric sampler for a wide range of dust types and particle sizes. The PDM averaging period will depend on the concentration of airborne dust, but is likely to be minutes rather than seconds.

The Respicon TM also showed good linearity compared with a reference IOM sampler for a wide range of dust types and particle sizes during laboratory trials, slightly overestimating the average inhalable airborne dust concentration. Temporal variations in inhalable airborne dust concentration measured using the Respicon TM also largely followed those of the modified PDM monitor. The main drawback of the Respicon TM is that it does not give a direct reading of airborne inhalable dust concentration; the readings can only be displayed after the measurement has been completed following gravimetric analysis of the filters.

Workplace measurements

Photometer-type direct-reading dust monitors consistently underestimated the measurement of inhalable airborne dust concentration, as was discovered in the laboratory tests. Likewise, their response changed significantly if the particle size varied throughout the measurement period. This resulted in inaccuracies when the photometer readings were scaled using average CFs determined from the ratio of reference IOM gravimetric sampler concentration to average photometer concentration.

Measurements made with a modified PDM during site visits showed that it performed well when used as a static inhalable dust monitor and agreed closely with the reference IOM gravimetric inhalable sampler. However, it was found to be susceptible to errors caused by movement making it unsuitable in its present form as a personal dust monitor. It was also found to be overly sensitive to large particles because of the configuration of the inlet. If the modified PDM could be designed to be less sensitive to movement and be fitted with a properly designed inhalable inlet, it could potentially be used to measure personal exposure to inhalable dust in real time.

As in the laboratory trials, the Respicon TM agreed closely with the reference IOM sampler, slightly overestimating the average inhalable airborne dust concentration.

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REFERENCES

CEN. (1993) Workplace atmospheres: size fraction definitions for measurement of airborne particles in the workplace. CEN standard EN 481. Brussels, Belgium: Comite Europe´en de Normalisation.

CEN. (2002) Workplace atmospheres: assessment of performance of instruments for measurement of airborne particle concentrations. CEN standard EN 13205. Brussels, Belgium: Comite Europe´en de Normalisation.

Higgins RJ, Dewell P. (1967) A gravimetric size selecting personal dust sampler. In Davies CN, editor. Inhaled particles and vapors II. Oxford: Pergamon Press. pp. 575–86.

HSE. (2000) Methods for the determination of hazardous substances. General methods for sampling and gravimetric analysis of respirable and inhalable dust. MDHS 14/3. Sudbury, UK: Health and Safety Executive Books.

ISO 12103-1. (1997) Test dust for filter evaluation. Part 1: Arizona Test Dust. Geneva, Switzerland: International Organization for Standardization (ISO).

Mark D, Vincent JH. (1986) A new personal sampler for airborne total dust in workplaces. Ann Occup Hyg; 30: 89–102.

Mark D, Vincent JH, Gibson H *et al.* (1985) Applications of closely-graded powders of fused alumina as test dusts for aerosol studies. J Aerosol Sci; 16: 125–31.

Niven RW. (1993) Aerodynamic particle size testing using a time-of-flight aerosol beam spectrometer. Pharm Technol; 17: 72–78.

Patashnick H, Meyer M, Rogers B. (2002). Tapered element oscillating microbalance technology. Mine ventilation, edited by Euler De Souza Proceedings of the North American/Ninth U.S. Mine Ventilation Symposium, Kingston, Ontario, Canada, 8–12 June 2002. pp. 625–631.

Rosén G, Andersson IM, Walsh PT *et al.* (2005) A review of video exposure monitoring as an occupational hygiene tool. Ann Occup Hyg; 49: 201–17.

Thorpe A. (2007) Assessment of personal direct-reading dust monitors for the measurement of airborne inhalable dust. Ann Occup Hyg; 51: 97–112.

Thorpe A, Walsh PT. (2007) Comparison of portable, real-time dust monitors sampling actively, with size-selective adaptors, and passively. Ann Occup Hyg; 51: 679–91.

Volkwein JC, Thimons E, Dunham D *et al.* (2004) Development and evaluation of a new personal dust monitor for underground mining applications. Proceedings of the 29th International Technical Conference on Coal Utilization & Fuel Systems, Clearwater Florida. Volume II, 18–22 April 2004. pp. 1355–75.

Willeke K, Baron PA. (2001) Aerosol measurement—principles, techniques and applications. New York, NY: Van Nostrand Reinhold.